MATHEMATICS

Applications and interpretation

Higher Level | Set D | Paper 1

Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

Question 1 [Maximum mark: 6]

A project costs 18 000 dollars now and returns 4000, 5000, 6000, 6000 and 7000 dollars at the ends of years 1 to 5 respectively. Its net present value at annual discount rate r is defined by $NPV(r) = -18000 + \sum_{t=1}^{5} C_t/(1+r)^t$. The internal rate of return is the value of r giving zero NPV.

(a) Calculate the net present value at a 6% discount rate. [2]

(b) Find the internal rate of return, as an annual percentage. [2]

(c) At a required return of 10%, decide whether the project has a positive NPV. Give one reason the calculation alone does not guarantee a profitable outcome. [2]

Question 2 [Maximum mark: 6]

The radius of a cylindrical tank is recorded as 2.40 m to the nearest 0.01 m. Its height is recorded as 5.2 m to the nearest 0.1 m. The nominal volume is calculated from these recorded values. Use the standard half-unit rounding intervals.

(a) State the lower and upper bounds for radius and height. [2]

(b) Find bounds for the true tank volume. [2]

(c) Find the largest possible percentage by which the nominal volume overstates the true volume. Use the true volume in the denominator. [2]

Question 3 [Maximum mark: 6]

A monthly energy bill consists of a fixed charge of 8 dollars, plus 0.18 dollars per kWh for the first 120 kWh and 0.32 dollars per kWh above 120. Monthly use X has the following distribution.

x / kWh	50	100	150	200
$P(X = x)$	0.1	0.3	0.4	0.2

(a) Find the bill for 150 kWh. [1]

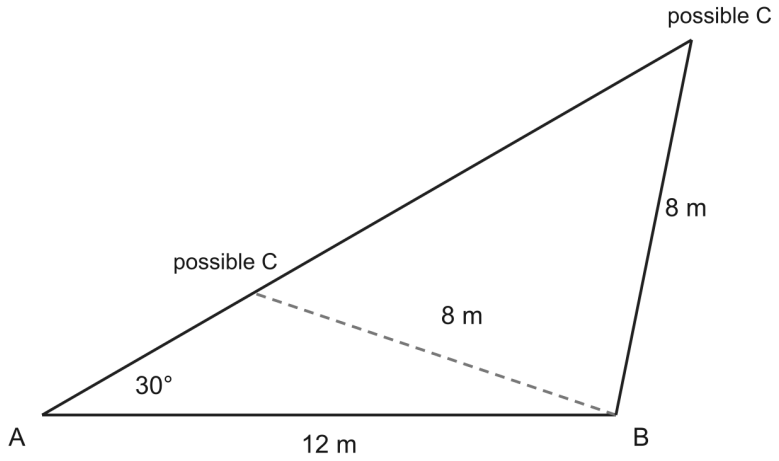
(b) Find the expected monthly bill. [3]

(c) Find the bill at the expected use $E(X)$ and compare it with part (b). Explain the difference. [2]

Question 4 | Question paper

Question 4 [Maximum mark: 6]

Survey points form a triangle ABC with $AB = 12$ m, $BC = 8$ m and $\angle BAC = 30°$. Both geometrically possible positions of C above AB are permitted initially.



Two possible triangles; not to scale

(a) Find the two possible values of $\angle ACB$. [2]

(b) Find the two possible lengths of AC. [3]

(c) The perpendicular projection of C onto the line AB lies strictly between A and B. Select the valid length of AC. [1]

Question 5 | Question paper

Question 5 [Maximum mark: 6]

A researcher records incident counts in 200 independent equal-duration intervals. The complete ungrouped counts sum to 400. Their grouped frequencies are

Count	0	1	2	≥ 3
Observed	20	58	62	60

Test the fit of a Poisson model at the 5% level, estimating its mean from these same observations.

(a) Estimate the Poisson mean. [1]

(b) Find all four expected frequencies. [2]

(c) State the degrees of freedom and find the test p-value. [2]

(d) State a conclusion in context. [1]

Question 6 | Question paper

Question 6 [Maximum mark: 7]

A crop response Y is modelled by $Y = 80 - Ae^{-kt}$, with $A, k > 0$ and t in weeks. The value 80 is a proposed saturation level. Data are

t	0	1	2	3	4
Y	18.1	31.9	44.3	51.4	58.9

Fit by least squares to the transformed linear relationship.

(a) State a linearized equation and find its fitted slope and intercept. [3]

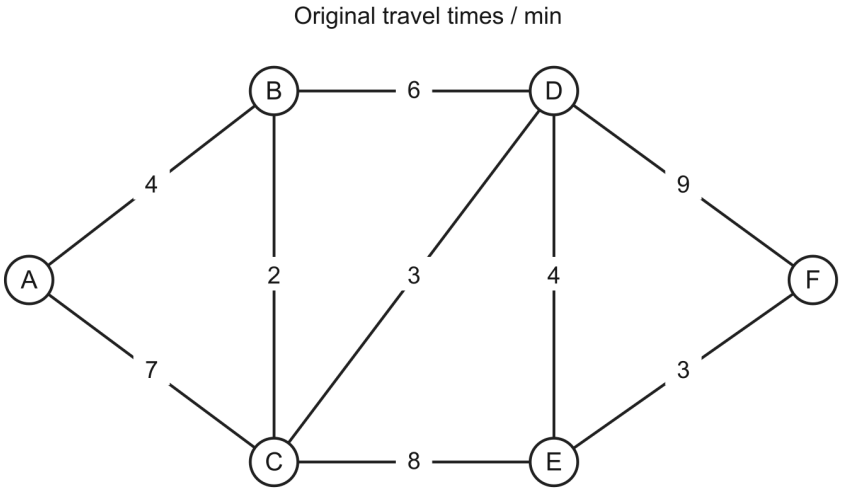
(b) Find A and k. [2]

(c) Find the fitted initial response and explain why the assumed saturation level cannot be established from these five early observations alone. [2]

Question 7 | Question paper

Question 7 [Maximum mark: 7]

The undirected graph shows travel times in minutes. A traveller goes from A to F. A proposed upgrade changes only the time on BD from 6 to x minutes, where $0 \leq x \leq 6$.



(a) Find a shortest original route from A to F and its time. [3]

(b) Find the values of x for which the upgrade gives a strictly shorter journey than the original minimum. [2]

(c) The upgrade must reduce the original minimum journey time by at least 20%. Find the largest permitted value of x. [2]

Question 8 [Maximum mark: 7]

Individual service times are independent, right-skewed, with population mean 18 minutes and standard deviation 12 minutes. For sample sizes greater than 30, use the central limit theorem to approximate the distribution of the sample mean.

(a) For a sample of 64 customers, estimate the probability that their mean service time exceeds 20 minutes. [2]

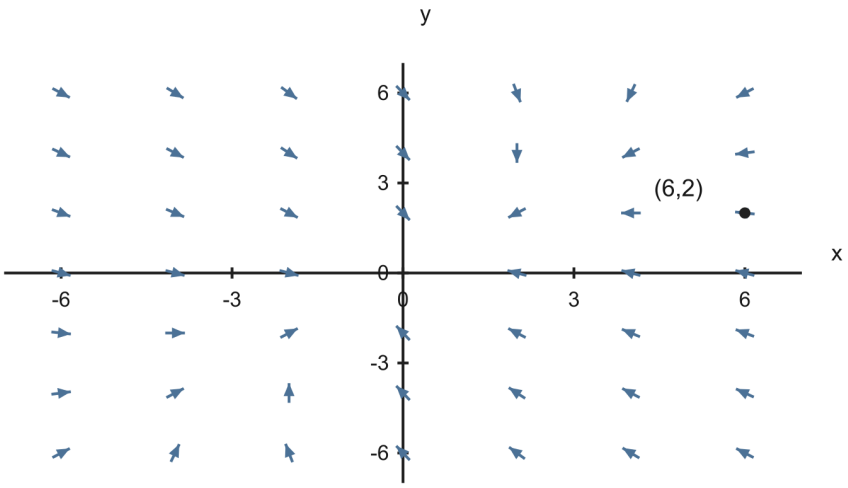
(b) For samples of 64, find a bound exceeded by only 1% of sample means. [2]

(c) Find the smallest integer sample size greater than 30 for which the approximate probability of a sample mean above 20 is at most 0.025. Check the neighbouring sample size. [3]

Question 9 | Question paper

Question 9 [Maximum mark: 7]

A linear model near a stable operating point is $dx/dt = -2x + y$, $dy/dt = x - 2y$. Time is measured in hours. Initially $(x, y) = (6, 2)$. The diagram shows the phase direction field.



(a) Find both eigenvalues and a corresponding eigenvector for each. [3]

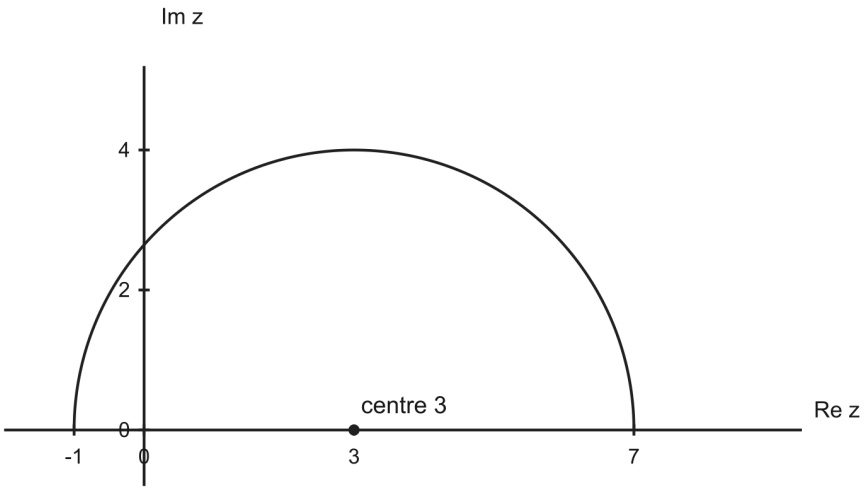
(b) Find $x(t)$ and $y(t)$ satisfying the initial condition. [3]

(c) State the line along which the trajectory approaches the origin, and justify it. [1]

Question 10 | Question paper

Question 10 [Maximum mark: 7]

The resultant phasor is $z = 3 + 4e^{i\theta}$, where $0 \leq \theta \leq \pi$. The diagram shows its locus. Arguments and θ are in radians.



(a) Describe the locus, giving its centre and radius. [2]

(b) Find θ when $\arg z = \pi/4$. [2]

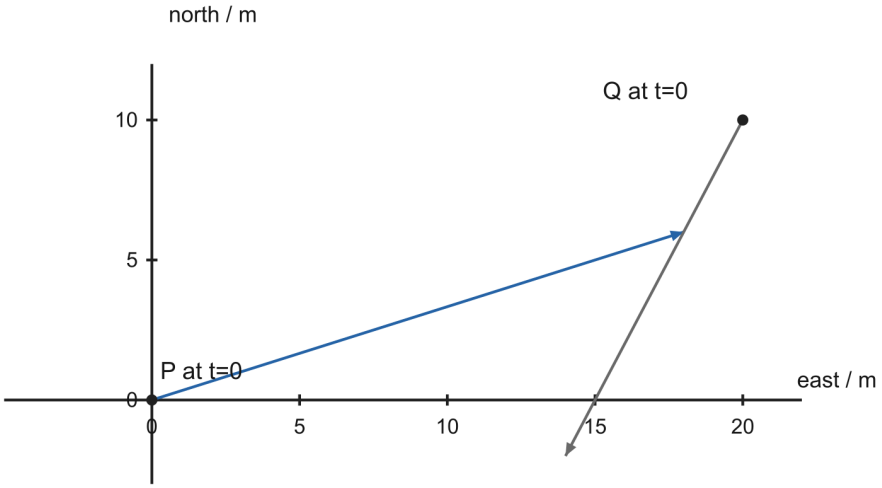
(c) Find the amplitude of this resultant. [2]

(d) Determine whether complete cancellation is possible for any permitted θ. [1]

Question 11 | Question paper

Question 11 [Maximum mark: 7]

Two inspection robots move in a horizontal plane for $0 \le t \le 6$ minutes. Their positions, in metres east and north, are $P(t) = (3t, t)$ and $Q(t) = (20 - t, 10 - 2t)$. The arrows show their paths, not simultaneous positions.



(a) Find the time of closest approach and the minimum separation. [3]

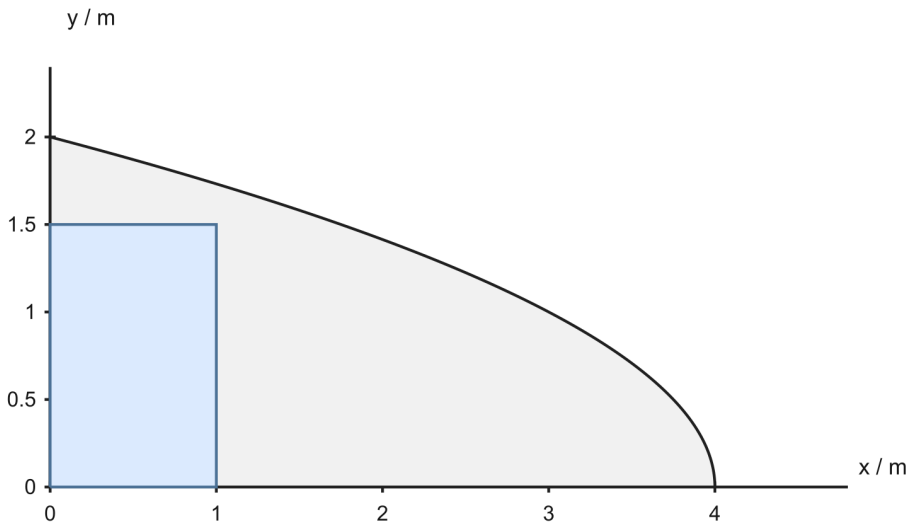
(b) Find the interval during which the robots are at most 5 m apart. [3]

(c) Explain whether the robots collide, despite the crossing paths in the diagram. [1]

Question 12 | Question paper

Question 12 [Maximum mark: 7]

The region $0 \leq y \leq 2, 0 \leq x \leq 4 - y^2$ is rotated about the y-axis to make a solid. All coordinates are in metres. A cylindrical hole of radius 1 m is drilled from the flat base $y = 0$ along the axis. The shaded rectangle shows the hole cross-section for depth 1.5 m.



(a) Find the volume of the original solid. [3]

(b) Find the remaining material volume after drilling to a depth of 1.5 m. [2]

(c) Find the greatest drilling depth possible before a radius-1 hole would break through the curved surface. [2]

Question 13 | Question paper

Question 13 [Maximum mark: 7]

A contamination rate is suspected to exceed 0.02. In a design for a one-tailed exact binomial test, n independent packets will be sampled and the null hypothesis will be rejected if at least two are contaminated. Choose the largest sample size whose Type I error probability is strictly below 0.05.

(a) State the null and alternative hypotheses for contamination probability p. [1]

(b) Find the largest permitted value of n, checking the next integer. [3]

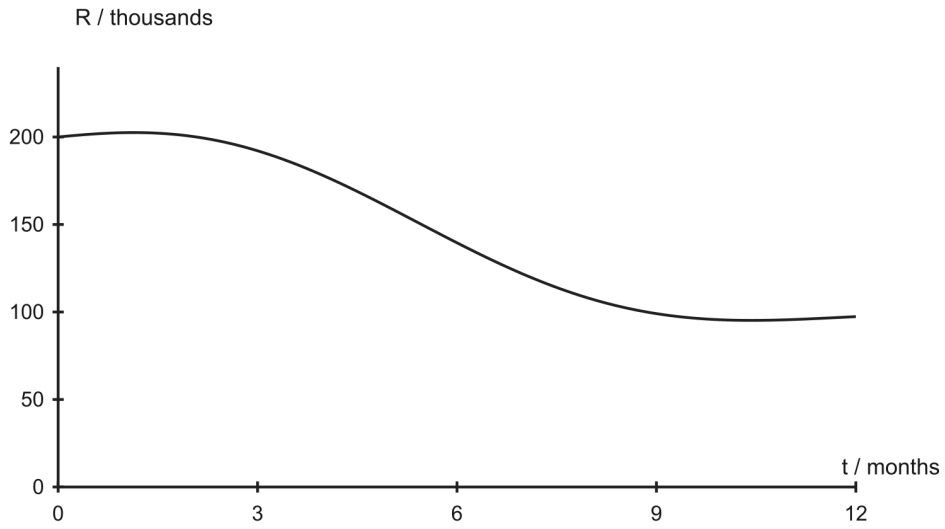
(c) For this sample size, find the power of the test when the true contamination probability is 0.12. [2]

(d) Explain why increasing the sample size without changing the rejection rule is not automatically permitted. [1]

Question 14 | Question paper

Question 14 [Maximum mark: 8]

The revenue forecast for a promotion held t months after a product launch is $R(t) = (200 + 30\sin(\pi t/6))e^{-0.06t}$ thousand dollars, for $0 \le t \le 12$. The seasonal term and declining overall interest are both part of the model. Angles are in radians.



(a) Find an expression for $R\prime(t)$. [2]

(b) Find all stationary times in the permitted interval. [3]

(c) Find the promotion time giving the global maximum forecast revenue and state that revenue. Justify using all candidates. [3]

Question 15 | Question paper

Question 15 [Maximum mark: 8]

Customers use an app A, website W or telephone P each week. Column state vectors use that order. The regular transition matrix is

$$T(q) = \begin{pmatrix} 0.7 & 0.1 & q \\ 0.2 & 0.7 & 0.2 \\ 0.1 & 0.2 & 0.8 - q \end{pmatrix}$$

, where $0.1 \leq q \leq 0.5$. The initial proportions are $(1/3, 1/3, 1/3)^T$. After one week, the app proportion is 0.40.

(a) Find q. [2]

(b) Find the long-run proportions for this value of q. [3]

(c) A new policy increases q to 0.5, so the bottom-right entry becomes 0.3; all constant entries remain unchanged. For 1000 customers, find the change in the long-run expected app count. [3]

Question 16 | Question paper

Question 16 [Maximum mark: 8]

The amount $M(t)$ of a substance in a filter, in grams, follows $dM/dt = -(0.1 + 0.02t)M$, with $M(0) = 80$, for $0 \leq t \leq 10$ hours. The removal rate is defined as the positive quantity $r(t) = -dM/dt$.

(a) Find an expression for $M(t)$. [3]

(b) Find the time when half the initial amount remains. [2]

(c) Find the time of maximum removal rate and the value of that rate, checking that it is the maximum on the model domain. [3]

End of question paper

MATHEMATICS

Applications and interpretation

Higher Level | Set D | Paper 1

Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 [Maximum mark: 6]

(a) [2 marks]

M1 Discount every return for its own number of years and subtract the initial cost.

A1 $NPV(0.06) = 5244.652$ dollars.

(b) [2 marks]

M1 Solve $NPV(r) = 0$ numerically for $r > 0$.

A1 The internal rate is 15.245771% per year.

(c) [2 marks]

A1 $NPV(0.10) = 2721.0138$ dollars, which is positive; the model meets the required return.

R1 Future returns are forecasts; costs, timing or cash flows may differ, so a positive model NPV does not remove commercial risk.

Question 2 [Maximum mark: 6]

(a) [2 marks]

A1 $2.395 \leq r < 2.405$ m.

A1 $5.15 \leq h < 5.25$ m.

(b) [2 marks]

M1 Volume $\pi r^2 h$ increases with each positive dimension.

A1 $92.804308 \leq V < 95.398015$ m^3, with the unrounded expressions $\pi(2.395)^2(5.15)$ and $\pi(2.405)^2(5.25)$.

(c) [2 marks]

M1 The nominal volume is 94.096983 m^3; the overstatement is greatest at the lower true-volume bound.

A1 $100(V_{\text{nom}} - V_{\text{min}})/V_{\text{min}} = 1.3929041\%$.

Question 3 [Maximum mark: 6]

(a) [1 marks]

A1 $8 + 0.18(120) + 0.32(30) = 39.20$ dollars.

(b) [3 marks]

M1 The bills at 50,100,150,200 kWh are 17.00,26.00,39.20,55.20 dollars.

M1 Weight those four bills by their probabilities.

A1 The expectation is 36.22 dollars.

(c) [2 marks]

A1 $E(X) = 135$ kWh and its bill is 34.40 dollars, below the expected bill.

R1 The tariff is not one linear rule across the full range; high-use months incur the higher marginal rate, so substituting the mean use does not give the mean bill.

Question 4 [Maximum mark: 6]

(a) [2 marks]

M1 $\sin C = 12 \sin 30^\circ / 8 = 0.75$.

A1 $C = 48.590378^\circ$ or 131.40962°.

(b) [3 marks]

M1 For each possible C, find B from the triangle angle sum.

M1 Apply the sine rule $AC = 8 \sin B / \sin 30^\circ$.

A1 The lengths are 5.1008022 m and 15.683807 m.

(c) [1 marks]

A1 Choose 5.1008022 m, whose projection is between 0 and 12 m. The larger length projects beyond B.

Question 5 [Maximum mark: 6]

(a) [1 marks]

A1 $\hat{\lambda} = 400/200 = 2$.

(b) [2 marks]

M1 Use Poisson probabilities with the fitted mean 2; pool all counts at least 3 into the last cell.

A1 The expected frequencies are $[27.067057, 54.134113, 54.134113, 64.664717]$.

(c) [2 marks]

A1 $df = 4 - 1 - 1 = 2$, subtracting one fitted parameter.

A1 $\chi^2 = 3.6006849$ and $p = 0.16524229$.

(d) [1 marks]

R1 $p = 0.16524229 > 0.05$, so do not reject the Poisson model. This is insufficient evidence of lack of fit, not proof that the model is true.

Question 6 [Maximum mark: 7]

(a) [3 marks]

M1 $\ln(80 - Y) = \ln A - kt$.

A1 The fitted slope is -0.26723697.

A1 The fitted intercept is 4.1298005.

(b) [2 marks]

A1 $A = e^{4.1298005} = 62.16552$.

A1 $k = 0.26723697 \text{ week}^{-1}$.

(c) [2 marks]

A1 $Y(0) = 80 - A = 17.83448$.

R1 The observations do not reach the long-run plateau. Other saturation levels/models may fit early data, so 80 remains a modelling assumption requiring later evidence.

Question 7 [Maximum mark: 7]

(a) [3 marks]

M1 Compare feasible route totals or use a consistent shortest-route method.

A1 A-B-C-D-E-F is a shortest route.

A1 Its time is 16 minutes.

(b) [2 marks]

M1 The upgraded route A-B-D-E-F takes $11 + x$ minutes; the old route still takes 16.

A1 A strict improvement occurs when $x < 5$. At $x = 5$ the two routes tie.

(c) [2 marks]

M1 Require $11 + x \leq 0.8(16) = 12.8$.

A1 The largest value is $x = 1.8$ minutes.

Question 8 | Worked solutions

Question 8 [Maximum mark: 7]

(a) [2 marks]

M1 The approximate sampling distribution is $\bar{X} \sim N(18, (12/8)^2)$.

A1 The probability is 0.09121122.

(b) [2 marks]

M1 $c = 18 + 1.5\Phi^{-1}(0.99)$.

A1 $c = 21.489522$ minutes.

(c) [3 marks]

M1 Require $(20 - 18)/(12/\sqrt{n}) \geq \Phi^{-1}(0.975)$.

A1 The smallest size is $n = 139$.

R1 The tails for $n = 138$ and $n = 139$ are 0.02512146 and 0.024708796 respectively, straddling 0.025.

Question 9 | Worked solutions

Question 9 [Maximum mark: 7]

(a) [3 marks]

M1 The characteristic equation is $(-2-\lambda)^2 - 1 = 0$.

A1 $\lambda = -1$ has eigenvector $(1, 1)^T$.

A1 $\lambda = -3$ has eigenvector $(1, -1)^T$.

(b) [3 marks]

M1 $\mathbf{s} = ae^{-t}(1, 1)^T + be^{-3t}(1, -1)^T$.

M1 At $t = 0$, $a + b = 6$, $a - b = 2$, so $a = 4, b = 2$.

A1 $x = 4e^{-t} + 2e^{-3t}$, $y = 4e^{-t} - 2e^{-3t}$.

(c) [1 marks]

R1 The e^{-t} term dominates the faster-decaying e^{-3t} term, so the trajectory approaches along $y = x$.

Question 10 | Worked solutions

Question 10 [Maximum mark: 7]

(a) [2 marks]

A1 It is the upper semicircle with centre $3 + 0i$.

A1 Its radius is 4, including both real-axis endpoints.

(b) [2 marks]

M1 Require $4\sin\theta = 3 + 4\cos\theta > 0$ and solve on $[0, \pi]$.

A1 The valid solution is $\theta = 1.344388$ rad.

(c) [2 marks]

M1 Use the modulus at the solution in part (b).

A1 $|z| = 5.5124853$.

(d) [1 marks]

R1 No: the smallest possible amplitude is $|4 - 3| = 1 > 0$, attained at $\theta = \pi$.

Question 11 | Worked solutions

Question 11 [Maximum mark: 7]

(a) [3 marks]

M1 The squared separation is $(4t-20)^2+(3t-10)^2=25t^2-220t+500$.

M1 Minimize on $0\le t\le 6$: $50t-220=0$.

A1 $t=4.4$ minutes and minimum distance 4 m.

(b) [3 marks]

M1 Require $25t^2-220t+500\le 25$.

A1 The boundary times are 3.8 and 5.0 minutes.

A1 The interval is $3.8\le t\le 5.0$ minutes.

(c) [1 marks]

R1 They do not collide: their minimum simultaneous separation is 4 m. Crossing geometric paths does not mean occupying the intersection at the same time.

Question 12 | Worked solutions

Question 12 [Maximum mark: 7]

(a) [3 marks]

M1 At height y, the disc radius is $x = 4 - y^2$.

M1 $V = \pi \int_0^2 (4 - y^2)^2 \, dy$.

A1 $V = 53.616515 \text{ m}^3$.

(b) [2 marks]

M1 The removed cylinder has volume $\pi(1)^2(1.5)$ and fits within the solid over that depth.

A1 The remaining volume is 48.904126 m^3.

(c) [2 marks]

M1 Require $4 - y^2 \geq 1$ throughout the hole; equality first occurs at $y = \sqrt{3}$.

A1 The maximum depth is 1.7320508 m.

Question 13 | Worked solutions

Question 13 [Maximum mark: 7]

(a) [1 marks]

A1 $H_0 : p = 0.02, H_1 : p > 0.02.$

(b) [3 marks]

M1 The Type I probability is $P_{0.02}(X \geq 2) = 1 - (0.98)^n - n(0.02)(0.98)^{n-1}$.

A1 The largest size is $n = 18$, with Type I probability 0.049508833.

R1 For $n = 19$ the probability is $0.05461595 > 0.05$.

(c) [2 marks]

M1 Calculate $P(X \geq 2)$ for $X \sim B(18, 0.12)$.

A1 The power is 0.65399768.

(d) [1 marks]

R1 With the threshold fixed at two, increasing n raises the null probability of rejection as well as power; a larger n can violate the Type I error limit.

Question 14 [Maximum mark: 8]

(a) [2 marks]

M1 Use the product and chain rules.

A1 $R\prime = e^{-0.06t}[5\pi\cos(\pi t/6) - 0.06(200 + 30\sin(\pi t/6))]$.

(b) [3 marks]

M1 Solve the bracketed expression equal to zero on $[0, 12]$.

A2 The times are $t = 1.136294$ and $t = 10.427899$ months.

(c) [3 marks]

M1 Compare endpoint and stationary values: $[200.0, 202.525072, 95.21302, 97.350451]$ thousand dollars at $[0, 1.136294, 10.427899, 12]$ months.

A1 The maximum occurs at $t = 1.136294$ months.

A1 The maximum forecast is 202.52507 thousand dollars.

Question 15 | Worked solutions

Question 15 [Maximum mark: 8]

(a) [2 marks]

M1 The app proportion after one step is $(0.7 + 0.1 + q)/3$.

A1 $q = 0.4$.

(b) [3 marks]

M1 Solve $T\boldsymbol{\pi} = \boldsymbol{\pi}$ with components summing to 1.

A2 The long-run vector is

$$\begin{pmatrix} 0.4 \\ 0.4 \\ 0.2 \end{pmatrix}$$

(c) [3 marks]

M1 Recompute the stationary state using $T(0.5)$.

A1 The new app probability is 0.425.

A1 The expected app count increases by $1000(0.425 - 0.4) = 25$ customers.

Question 16 [Maximum mark: 8]

(a) [3 marks]

M1 Separate: $dM/M = -(0.1 + 0.02t)\,dt$.

M1 $\ln M = -0.1t - 0.01t^2 + C$ and use $M(0) = 80$.

A1 $M = 80e^{-0.1t - 0.01t^2}$.

(b) [2 marks]

M1 Solve $0.1t + 0.01t^2 = \ln 2$ on $[0, 10]$.

A1 The time is $t = 4.7115765$ hours.

(c) [3 marks]

M1 $r\prime = M[0.02 - (0.1 + 0.02t)^2]$; it changes from positive to negative when $0.1 + 0.02t = \sqrt{0.02}$.

A1 The maximizing time is $t = 2.0710678$ h, within the permitted interval.

A1 The maximum removal rate is 8.811125 g h^{-1}; it exceeds both endpoint rates.

End of worked solution book